CGIAR
SCIENCE FOR A FOOD-SECURE FUTURE

POLICY INNOVATIONS

Mapping the risky returns to smallholder fertilizer investments in Africa

December 22, 2025

CGIAR Policy Innovations Program Report

Bisrat Gebrekidan[1], Jordan Chamberlin[1], Maxwell Mkondwa[1]; Joao Vasco Silva[2] and Robert Hijmans[3]

1 CIMMYT
2 Uni-Hohenheim
3 UC Davis

Acknowledgements: This work was carried out under the CGIAR Science Program on Policy Innovations, CGIAR Science Program on Sustainable Farming and through the Guiding Acid Soil Management Investments in Africa (GAIA) project (Grant no: INV-029117), supported by the Bill & Melinda Gates Foundation (BMGF). We thank all funders who supported this research through their contributions to the CGIAR Trust Fund.

Citation: Gebrekidan, B., J. Chamberlin, M. Mkondwa, J. Silva, & R. Hijimans (2025). Mapping the risky returns to smallholder fertilizer investments in Africa. CGIAR Policy Innovation Program Report.

Disclaimer: This report has not been peer reviewed. Any opinions stated herein are those of the author(s) and do not necessarily reflect the policies or opinions of CIMMYT, CGIAR, donors, or partners. This publication is copyrighted by CIMMYT. It is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License. To view this license, visit https://creativecommons.org/licenses/by/4.0. Unless otherwise noted, you are free to share (copy and redistribute the material in any medium or format) and adapt (remix, transform, and build upon the material) for any purpose, under the following conditions:

Attribution: The work must be attributed, but not in any way that suggests endorsement by CIMMYT or the author(s).

Contents

1. INTRODUCTION **1**

2. MODELING FRAMEWORK **3**

2.1 Overview and modeling logic 3
2.2 Decision problem and stochastic profit representation 5
2.3 Modeling spatially explicit input and output prices 5
2.4 Model-agnostic explanation and plausibility validation 6
2.5 Output price formation and volatility 6
2.6 Modeling crop failure 7
2.7 Monte Carlo simulation of profit distributions 9

3. DATA **9**

4. RESULTS **13**

4.1 Spatial patterns in predicted maize and nitrogen price surfaces 13
4.2 Spatial heterogeneity in yield response to nitrogen 13
4.3 Spatial distribution of predicted crop-loss probability 14
4.4 Joint spatial dependence from the spatial copula 15
4.5 Probability of loss and downside profit risk 17
4.6 Profit–risk quadrants and spatial targeting implications 19

5. DISCUSSIONS AND CONCLUSIONS **22**

6. LIMITATIONS AND OUTLOOK **23**

REFERENCES **25**

Abstract

Low and inconsistent use of inorganic fertilizer continues to constrain agricultural productivity growth in sub-Saharan Africa, despite strong agronomic evidence of yield responsiveness in maize-based systems. A central challenge is that fertilizer decisions are made under substantial production and market uncertainty, yet most policy targeting and advisory tools rely on average profitability metrics that mask exposure to downside risk. This paper develops a spatially explicit, *ex ante* framework to support risk-informed fertilizer policy and investment decisions under rainfed smallholder conditions. We integrate georeferenced maize fertilizer trial responses with spatial data on soils, long-run rainfall variability, remotely sensed vegetation, and locally differentiated input and output prices to characterize the full distributions of fertilizer returns across space. Production risk, price volatility, and crop failure are modeled explicitly, including their joint dependence through spatial rainfall and price relationships. The results show that areas with high expected returns frequently face a substantial probability of financial loss, driven by the interaction of climatic variability, market volatility, and catastrophic production risk. Accounting for these risks identifies spatial “risk traps” where fertilizer is technically productive but economically unattractive for risk-averse farmers. The framework provides guidance for the spatial targeting of fertilizer subsidies, extension, insurance, and complementary market interventions, helping policymakers and investors prioritize locations where fertilizer use can be both profitable and resilient, and tailor risk-mitigation instruments where uncertainty remains binding.

Keywords: Fertilizer profitability; Production risk; Market volatility; Crop failure; Spatial targeting; Rainfed agriculture; Smallholder farming; Spatial copulas; Risk-adjusted returns; Decision-making under uncertainty; Sub-Saharan Africa

1.Introduction

The limited use of inorganic fertilizer in smallholder farming systems across sub-Saharan Africa has been a persistent concern for agricultural policy for several decades. Both the extensive margin, whether farmers use fertilizer at all, and the intensive margin, the quantities they apply, remain far below levels widely considered necessary for sustained agricultural productivity growth (Burke et al., 2019; Jayne et al., 2018). This challenge has been repeatedly emphasized in high-level policy processes, most notably the Abuja Fertilizer Summit in 2008 and, more recently, the African Union's Fertilizer and Soil Health Summit held in Nairobi in 2024. Together, these initiatives reflect a broad consensus that increasing fertilizer use is central to improving soil health, narrowing yield gaps, and strengthening food security in the region's predominantly rainfed smallholder farming systems.A key explanation put forward in empirical studies for the consistently low use of fertilizers is that the economic returns are often lower and more variable than those assumed in policy design (Darko et al., 2024, Ragasa et al., 2025, Koussoube et al., 2016, Liverpool-Tasie et al., 2017). Micro-level evidence shows that the average returns used to rationalize fertilizer promotion frequently overstate the returns realized by farmers under actual production and market conditions (Suri 2011; Beaman et al. 2013; Holden 2018; Golin and Udry 2021; Suri and Udry 2022; Burke et al., 2016). Heterogeneity in soils, climate, and market access implies that fertilizer profitability varies sharply across space (Bonilla-Cedrez et al. 2021), and that for many farmers, observed returns fall below levels that would justify adoption given liquidity constraints and competing uses of scarce capital.

Recent advances in spatially explicit modeling have substantially improved our ability to characterize heterogeneity. By combining gridded soil data, weather information, and spatially varying input and output prices, a growing body of literature has generated fine-scale estimates of agronomic responses and financial returns to fertilizer use across diverse agroecological and market contexts (Palmas and Chamberlin 2020; Bonilla-Cedrez et al. 2021; McCullough et al. 2022). These approaches represent a major methodological advance over earlier, more aggregated analyses, enabling policymakers to move beyond blanket recommendations toward geographically targeted fertilizer promotion strategies and complementary investments, such as infrastructure improvements or smart subsidies, aimed at relaxing local profitability constraints in the long term.

However, an important dimension of fertilizer profitability remains insufficiently articulated in these spatial frameworks: the fact that returns to fertilizer are inherently uncertain and that this uncertainty itself varies systematically across space (Palmas and Chamberlin, 2020). Most existing models focus on expected or average returns, implicitly treating fertilizer profitability as a deterministic outcome conditional on location-specific characteristics (Assefa et al. 2021). Yet for smallholder farmers operating in rainfed systems, fertilizer investments are made under profound uncertainty (Druilhe et al., 2012). Rainfall outcomes are stochastic, prices fluctuate substantially, and formal mechanisms for insurance or consumption smoothing are largely absent (Abay et al., 2025; Krah et al., 2019). In such settings, farmers do not evaluate fertilizer based on expected profit alone, but rather on the full distribution of possible outcomes, with particular attention to the likelihood and severity of losses.

This distinction is central to understanding fertilizer adoption behavior. A substantial literature in development economics demonstrates that uncertain financial returns can strongly suppress productive investment by credit-constrained and risk-averse households (Dercon and Christiaensen 2011; De Brauw and Eozenou 2014; Emerick et al. 2016). When the downside risk is high, households may rationally avoid investments that appear profitable on average but expose them to unacceptable losses in adverse states of the world. Ignoring this dimension of risk therefore risks overstating the attractiveness of fertilizer use and misdirecting

policy efforts toward interventions that raise mean returns without addressing the underlying exposure to loss.

Recent work on rainfall-driven profitability risk in SSA maize systems further reinforces this point by emphasizing that rainfall variability translates into financial risk only through its interaction with markets, costs and management practices (McCullough et al., 2022). Yield shocks induced by rainfall variability affect revenues, but the welfare consequences depend critically on output prices, which may be volatile, spatially segmented, and only weakly responsive to local scarcity.

In this study, we address a key gap in the existing literature by developing an empirically grounded, spatially explicit framework for evaluating uncertain agronomic and economic returns to fertilizer investments. Rather than treating fertilizer profitability as a single expected value, we modeled it as a random variable whose distribution reflects location-specific uncertainty in production outcomes and market conditions. Using historical information on rainfall outcomes, we simulated a distribution of plausible future realizations of fertilizer-induced yield responses, conditional on local soils and climate. These simulated production outcomes were then translated into distributions of economic returns, yielding first-order estimates of expected profit, probability of loss, and downside risk faced by farmers at each location.

Our approach builds directly on recent spatial models of fertilizer response and profitability while extending them in a critical and policy-relevant direction. A growing body of work has shown that even when agronomic responses are strong, local realizations of input and output prices often undermine the profitability of fertilizer use (Palmas and Chamberlin 2020; Bonilla Cedrez et al. 2021; McCullough et al. 2022). Among these contributions, McCullough et al. (2022) developed a climate risk-aware framework that combines experimental maize trial data with spatially resolved soil and climate information to generate *ex ante* distributions of fertilizer returns under stochastic weather realizations across Sub-Saharan Africa. Similar to that study, we treat fertilizer use as an investment decision made prior to the realization of growing season conditions and evaluate profitability using full outcome distributions rather than mean responses alone. We extend this framework in three key ways. First, rather than focusing exclusively on production risk arising from climate variability, we model fertilizer profitability as the joint outcome of biophysical and market risk, allowing rainfall and price realizations to interact and co-move spatially. Second, we explicitly model crop failure as a discrete risk, distinguishing between partial yield reductions and catastrophic losses that result in near-zero output, an economically striking feature of highly variable rainfed systems that is typically not captured in agronomic trials and the earlier spatial profitability analyses derived from them. Third, rather than classifying locations solely based on robustness or profitability thresholds, we recover spatially explicit distributions of net returns and downside risk measures that reveal localized risk traps in which fertilizer is agronomically responsive yet financially unattractive in the absence of complementary risk mitigation instruments. Taken together, these extensions generalize existing climate-aware profitability frameworks into an integrated approach to risk-adjusted fertilizer investment targeting that more accurately reflects the compound and asymmetric risks faced by smallholder farmers.

Therefore, our contribution is not to identify new sources of heterogeneity in average returns but to show how spatially varying investment risk fundamentally alters the assessment of fertilizer profitability. We demonstrate that regions characterized by high expected returns may simultaneously exhibit substantial downside risk, rendering fertilizer adoption unattractive for risk-averse farmers in the absence of risk-sharing mechanisms, such as insurance. By jointly accounting for rainfall and market risk within a unified, spatially explicit framework, we provide a more realistic basis for understanding fertilizer adoption behavior and for designing policies that address not only low average returns but also high exposure to loss.

The remainder of the report is structured as follows. Section 2 introduces the modeling framework, outlining the decision problem, the representation of stochastic profits, and the

integration of production, market, and crop failure risk. Section 3 describes the data sources and preprocessing steps used to construct spatially explicit yield, price, and risk surfaces. Section 4 presents the main results, focusing on spatial heterogeneity in fertilizer profitability, downside risk, and the joint distribution of returns under climate and market uncertainty. Section 5 discusses and concludes the implications of these findings for fertilizer targeting and policy design. Section 6 presents discussion of limitations, validation considerations, and directions for future work.

2. Modeling framework

2.1 Overview and modeling logic

The objective of the empirical strategy is to recover the ex-ante distribution of net returns to fertilizer use faced by smallholder farmers under joint climate and market uncertainties. Rather than estimating average treatment effects or conditional mean responses, the framework was designed to approximate the full probability distribution of profits associated with fertilizer application at a given location. This requires explicitly modeling three distinct but interacting components: (i) the conditional yield response to fertilizer under varying biophysical conditions, (ii) the stochastic processes governing rainfall and output prices, and (iii) the dependence structure linking climate and market realization.

A key feature of this approach is the separation of marginal processes from their dependence structure. Agronomic trial data provide information on how yields respond to fertilizer, conditional on soils and weather, but do not identify how rainfall and prices co-evolve. Conversely, long-run rainfall and price series identify the marginal distributions of these variables but not their causal effects on the yield. By estimating each marginal process separately and subsequently coupling them through a flexible dependence structure, we avoid imposing strong parametric assumptions while preserving the empirical realism.

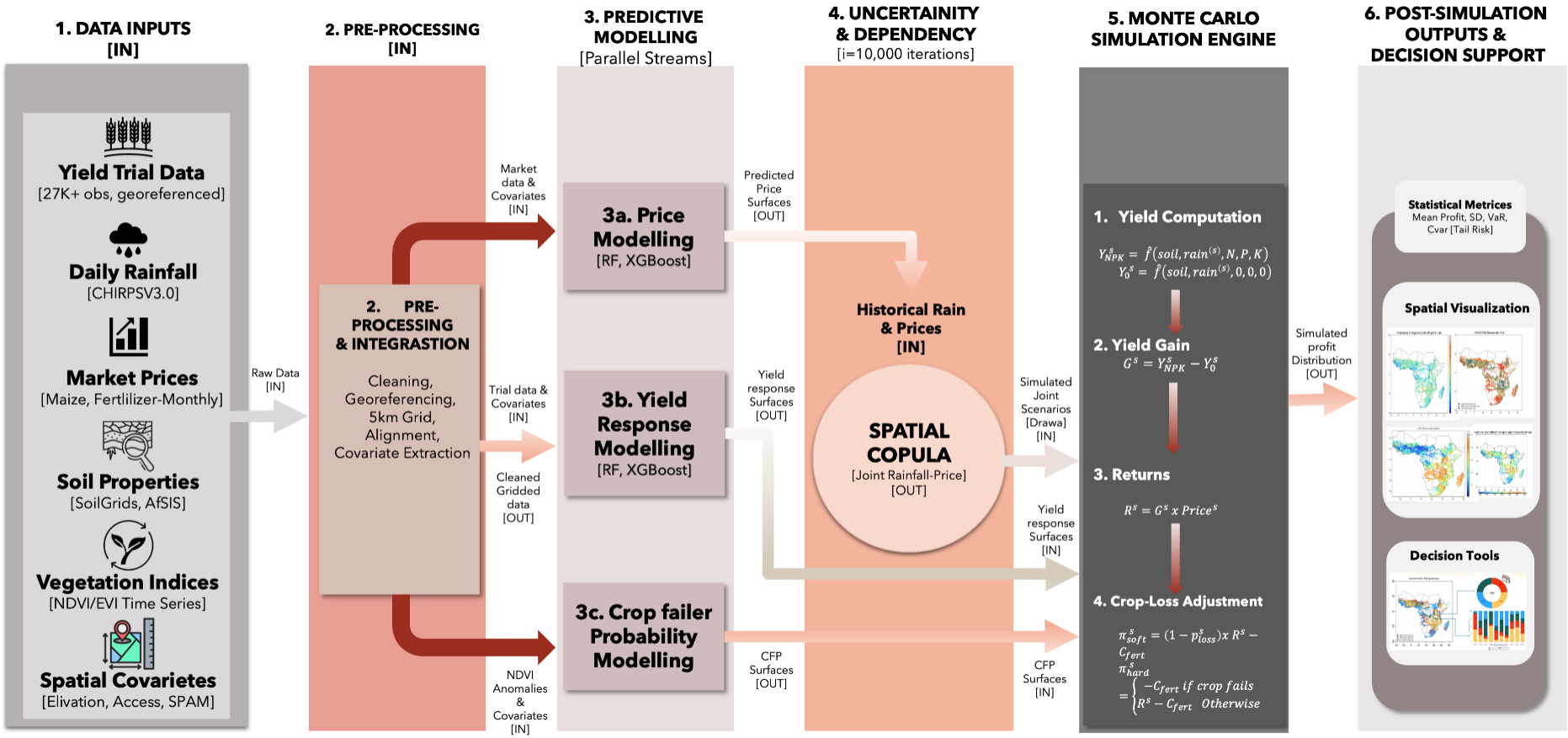

Figure 1:Integrated modeling pipeline linking agronomic data, climate and market information, and stochastic simulation to generate spatial profit and risk surfaces

2.2 Decision problem and stochastic profit representation

We frame fertilizer use as an *ex-ante* investment decision made before the realization of weather, prices, and crop outcomes. For a representative location i, the net profit from applying a fixed fertilizer package $\mathbf{f} = (N, P, K)$ in season t is given by

$$\pi_{it}(\mathbf{f}) = p_{it}[Y_{it}(\mathbf{f}) - Y_{it}(\mathbf{0})] - C_i(\mathbf{f}), \qquad (2.1)$$

where p_{it} denotes the realized output price, $Y_{it}(\mathbf{f})$ and $Y_{it}(\mathbf{0})$ denote fertilized and unfertilized yields, respectively, and $C_i(\mathbf{f})$ is the full economic cost of fertilizer. From the farmer's perspective, $\pi_{it}(\mathbf{f})$ is a random variable whose distribution is induced by uncertainty in rainfall, prices, and production outcomes.

Our objective is to recover the location-specific distribution of profits,

$$F_{\pi_i}(\pi) = \Pr!\,(\pi_{it}(\mathbf{f}) \leq \pi), \qquad (2.2)$$

rather than relying on point estimates, such as expected profit or internal rates of return. This distribution provides the basis for evaluating the downside risk, probability of loss, and other welfare-relevant metrics.

3.2 Yield response as a conditional prediction problem

The yield response to fertilizer was modeled as a conditional expectation function that maps agronomic inputs and realized environmental conditions to the expected output. Let $\mathbf{X}_{it} = (N_i, P_i, K_i, \mathbf{S}_i, R_{it})$ denote fertilizer rates, soil properties, and seasonal rainfall. We express yields as

$$Y_{it} = m(\mathbf{X}_{it}) + \varepsilon_{it}, \qquad (2.3)$$

where $m(\cdot) = \mathbb{E}[Y_{it} \mid \mathbf{X}_{it}]$ is an unknown response surface and ε_{it} captures idiosyncratic shocks and measurement error.

We approximate $m(\cdot)$ using tree-based ensemble learning, which allows for flexible nonlinear responses and high-order interactions without imposing parametric structure. These models are used strictly in a predictive sense to recover conditional mean responses within the support of the data, rather than to estimate structural parameters.

To propagate uncertainty into the profit distribution, we explicitly reintroduce stochastic error. For each simulation draw s, yields are generated as

$$\tilde{Y}_{it}^{(s)} = \widehat{m}\left(\mathbf{X}_{it}^{(s)}\right) + \eta_{it}^{(s)}, \qquad \eta_{it}^{(s)} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad (2.4)$$

where σ_ε^2 is calibrated to out-of-sample prediction error. Fertilizer-induced yield gains are computed by predicting fertilized and unfertilized counterfactuals under identical environmental conditions.

2.3 Modeling spatially explicit input and output prices

Profitability depends critically on both output prices received and input prices paid. We therefore model maize prices and fertilizer prices as spatially explicit stochastic processes, treated symmetrically with yield response.

Let p_{it}^y denote the farm-gate maize price and p_{it}^n the unit nitrogen price at location i. Fertilizer cost is given by

$$C_i(f) = p_{it}^n N_i + p_{it}^p P_i + p_{it}^k K_i + \tau_i, \qquad (2.5)$$

where τ_i captures location-specific transport and handling costs. Using cleaned and deflated market price data, we construct spatial price surfaces by estimating conditional expectations of prices as functions of market accessibility and spatial proximity. Flexible tree-based models are used to capture nonlinear gradients between integrated and remote markets.

As with yields, predicted prices serve as conditional means rather than realized outcomes. Location-specific empirical price distributions are used to generate stochastic price realizations in simulation, preserving observed volatility, skewness, and tail behavior. This ensures that market risk enters the profit distribution explicitly rather than being absorbed into average prices.

2.4 Model-agnostic explanation and plausibility validation

Because the framework relies on flexible, nonparametric prediction models, high predictive accuracy alone is insufficient to guarantee that learned relationships are economically or agronomically meaningful. Prior to their use in stochastic simulation, all machine-learning components are therefore subjected to a common set of model-agnostic explanation and plausibility checks.

These diagnostics characterize how predictions vary with key covariates and their interactions without reference to model internals. They are used to verify that learned response surfaces exhibit theoretically consistent patterns,such as diminishing marginal effects, sensible interaction between environmental stressors and inputs, and monotonic responses to risk factors,within the empirical support of the data. Importantly, explanations are evaluated locally rather than relying on global summaries, so that predictive structure remains plausible across heterogeneous soil, climate, and market environments.

Only models that satisfy these plausibility criteria are retained for simulation. This step is critical because implausible predictive patterns at the first stage would otherwise be mechanically propagated and amplified in downstream estimates of investment risk.

2.5 Output price formation and volatility

Output prices are modeled as stochastic and spatially heterogeneous. Let P_{it} denote the farm-gate price faced by farmers at location i in season t. Using historical market observations, we estimate location-specific empirical cumulative distribution functions $F_{P_i}(\cdot)$, which preserve observed volatility, skewness, and tail behavior.

Simulated prices are drawn as

$$\tilde{p}_{it}^{(s)} = F_{P_i}^{-1}\left(U_{it}^{P,(s)}\right), \qquad U_{it}^{P,(s)} \sim \mathcal{U}(0,1), \qquad (2.6)$$

providing that price uncertainty entering the profit calculation reflects local market conditions rather than national averages or parametric assumptions.

Rainfall and prices are not assumed to be independent. To model their joint behavior, we use a spatial Gaussian copula. Let R_{it} denote seasonal rainfall with marginal distribution $F_{R_i}(\cdot)$. We introduce latent Gaussian fields Z_{iR} and Z_{iP} such that

$$(Z_{iR}, Z_{iP}) \sim \mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\right), \qquad (2.7)$$

where ρ is the cross-correlation parameter. These latent variables are transformed to uniforms via the Gaussian CDF and mapped to empirical margins via inverse CDF mapping:

$$\tilde{R}_{it}^{(s)} = F_{R_i}^{-1}\left(\Phi\left(Z_{iR}^{(s)}\right)\right), \qquad \tilde{p}_{it}^{(s)} = F_{P_i}^{-1}\left(\Phi\left(Z_{iP}^{(s)}\right)\right). \qquad (2.8)$$

This construction preserves empirical rainfall and price distributions while allowing adverse rainfall and unfavorable price realizations to co-occur, capturing compound shocks relevant for smallholder decision-making.

2.6 Modeling crop failure

A key contribution of this study is the explicit treatment of crop failure as a discrete stochastic outcome that is conceptually distinct from continuous yield variability. In many spatial analyses of fertilizer profitability, extreme downside outcomes are implicitly represented as the lower tail of predicted yield distributions. This approach conflates moderate yield shortfalls with catastrophic losses and fails to capture the structural discontinuity faced by smallholders when crops fail almost entirely due to extreme climatic or biotic stress. In rainfed smallholder systems, such failure events imply outcomes that are qualitatively different from low-yield realizations, often resulting in near-total loss of the fertilizer investment. We therefore model crop failure as a separate probabilistic process that operates alongside, but independently from, the yield response function.

Let $F_{i,t} \in \{0,1\}$ denote an indicator of crop failure at location i in season t, where $F_{i,t} = 1$ corresponds to catastrophic loss and $F_{i,t} = 0$ to a productive outcome. We assume that failure arises from a latent stress process driven by extreme weather realizations and local biophysical conditions, and formalize this as

$$F_{i,t} = \mathbb{1}\big(g\big(\mathbf{W}_{i,t}, \mathbf{W}_{i,t-1}, \mathbf{X}_i\big) + \varepsilon_{i,t} > 0\big), \qquad (2.9)$$

where $\mathbf{W}_{i,t}$ denotes contemporaneous weather conditions (including seasonal rainfall levels and anomalies), $\mathbf{W}_{i,t-1}$ captures lagged climatic stress that may affect crop establishment, $\mathbf{X}_i$ represents time-invariant soil characteristics, and $\varepsilon_{i,t}$ is an unobserved shock. The function $g(\cdot)$ is unknown and potentially highly nonlinear, reflecting threshold behavior and interactions between moisture stress and soil buffering capacity.

We implement two alternative empirical strategies to approximate this failure process. These approaches are not combined at this stage of the analysis; rather, they represent distinct modeling options with different data requirements and spatial coverage.

Survey-based crop failure model

The first approach estimates crop failure probabilities directly from household survey data that report plot-level crop loss outcomes. These data provide the most direct empirical evidence on realized catastrophic losses, but they are spatially sparse and currently available for a limited set of countries and years. As such, this approach serves primarily as a benchmark and proof of concept, rather than as a continent-wide implementation.

Using georeferenced survey observations, we link reported crop failure indicators to realized seasonal rainfall, rainfall anomalies, lagged rainfall conditions, and soil characteristics. We estimate failure probabilities as

$$\hat{p}_{i,t}^{\text{survey}} = \Pr\big(F_{i,t} = 1 \mid \mathbf{W}_{i,t}, \mathbf{W}_{i,t-1}, \mathbf{X}_i\big), \qquad (2.10)$$

approximating the unknown function $g(\cdot)$ using a probabilistic random forest classifier. This approach accommodates nonlinear responses and interaction effects without imposing parametric structure, which is particularly important given the threshold-like nature of catastrophic loss. As a robustness check and for interpretability, we also estimate a probit specification based on the same covariates. Model performance is assessed using out-of-sample discrimination and calibration diagnostics to ensure that predicted probabilities correspond to observed frequencies of reported loss.

Importantly, fertilizer application rates are not included as predictors in the failure model. This reflects the conceptual distinction that fertilizer primarily affects yield conditional on crop survival, whereas catastrophic failure is driven by extreme stress conditions under which marginal nutrient response is largely irrelevant.

While the survey-based approach provides high-confidence identification of failure mechanisms where data are available, its limited geographic coverage precludes its direct use for large-scale spatial simulation at present. Expanding survey coverage to additional countries and seasons is therefore a priority for future work.

NDVI-based crop failure model

The second approach relies on remotely sensed vegetation indices to estimate crop failure probabilities in a spatially continuous manner. This approach is designed to be scalable across regions and time periods where survey-based loss data are unavailable and constitutes the primary implementation used in the simulation analysis.

We use end-of-season NDVI (or EVI) anomalies, expressed as standardized deviations from long-run local means, as indicators of extreme crop stress. Rather than imposing deterministic thresholds or mapping NDVI anomalies directly to failure probabilities, we treat vegetation stress as an explanatory variable in a probabilistic failure model. Specifically, we estimate

$$\Pr\left(F_{i,t} = 1 \mid Z_{i,t}^{\text{NDVI}}, Z_{i,t}^{\text{rain}}, \mathbf{X}_i\right), \qquad (2.11)$$

where $Z_{i,t}^{\text{NDVI}}$ denotes the vegetation anomaly and $Z_{i,t}^{\text{rain}}$ captures contemporaneous rainfall anomalies. This formulation recognizes that identical NDVI anomalies may correspond to different failure risks depending on the rainfall context and soil conditions, and that NDVI is an imperfect but informative signal of the underlying crop stress.

We estimate this relationship using flexible nonparametric models, including generalized additive models and random forest regressors, allowing failure probabilities to vary smoothly with stress intensity while accommodating interactions with soil characteristics. Model-agnostic explanation diagnostics are used to verify that estimated relationships exhibit agronomically plausible patterns, such as sharply increasing failure probabilities under extreme negative vegetation anomalies and limited sensitivity under favorable conditions.

Role of Crop Failure in Stochastic Return Simulations

Regardless of the empirical implementation used, the estimated crop failure probabilities enter the stochastic return framework as a discrete downside-risk channel. For each Monte Carlo draw at location i, rainfall realizations sampled from the spatial copula are used consistently to generate yield predictions, price realizations, and crop failure probabilities. A Bernoulli draw based on the predicted failure probability determines whether the simulated season results in catastrophic loss or a productive outcome.

If $F_{i,t} = 1$, profits are adjusted to reflect crop failure in one of two ways, corresponding to alternative representations of downside risk. In the *hard failure* specification, realized profits are set equal to the negative fertilizer investment cost, $-C_i(\mathbf{f})$, reflecting near-total loss of the fertilizer investment when catastrophic failure occurs. If $F_{i,t} = 0$, profits are computed from simulated yield gains and price realizations in the standard way. This formulation introduces a discrete mass point at extreme losses in the profit distribution, explicitly representing catastrophic outcomes that cannot be captured by continuous yield variation alone. In addition, we compute a *soft failure* or risk-adjusted profit measure, defined as expected profit conditional on failure risk, in which simulated profits are down-weighted by the predicted probability of crop failure, $\left(1 - \Pr\left(F_{i,t} = 1\right)\right)$. This representation captures environments where failure risk is elevated but not necessarily realized in a given season and provides a

smoother measure of expected downside exposure. Together, these two representations allow catastrophic losses to enter the profit distribution both as discrete events and as probabilistic risk, establishing that risk metrics such as the probability of loss, Value at Risk, and Conditional Value at Risk reflect both gradual yield variability and exposure to rare but severe failure events. This dual treatment provides a more complete characterization of the investment risk faced by smallholder farmers under rainfed conditions.

2.7 Monte Carlo simulation of profit distributions

All stochastic components are combined within a Monte Carlo framework. For each location, we simulate a large number of seasons, drawing joint rainfall–price realizations, predicting yields, evaluating crop failure, and computing profits as

$$\pi_{it}^{(s)} = \tilde{p}_{it}^{(s)} \cdot \Delta Y_{it}^{(s)} - C_i(\mathbf{f}). \qquad (2.12)$$

The resulting empirical distribution approximates $F_{\pi_i}(\cdot)$, from which expected profits, probabilities of loss, Value at Risk, and Conditional Value at Risk are computed.

All results are interpreted in an ex-ante investment framework. Fertilizer application decisions are fixed prior to uncertainty realization, while rainfall, prices, and crop outcomes are realized stochastically. This aligns the analysis with the farmer's decision problem and avoids conflating realized outcomes with expected investment performance.

3. Data

We compile a spatially explicit, multi-source dataset covering climate, soils, crop distribution, market prices, and crop outcomes across sub-Saharan Africa (SSA). All datasets are harmonized to a common spatial grid and temporal reference to support integrated modeling and simulation of agricultural profitability under uncertainty.

Yield response to fertilizer is estimated using a harmonized database of georeferenced, researcher-managed maize fertilizer trials conducted across multiple countries in Sub-Saharan Africa over the past three decades and harmonized using the Carob project workflow and associated Rpackage (Hijmans et al., 2023). The dataset includes trials in which nitrogen, phosphorus, and potassium application rates were experimentally varied, allowing identification of crop response under heterogeneous biophysical conditions. The final dataset comprises more than 27,000 plot-level observations, spanning a wide range of agroecological zones, soil types, and management environments.

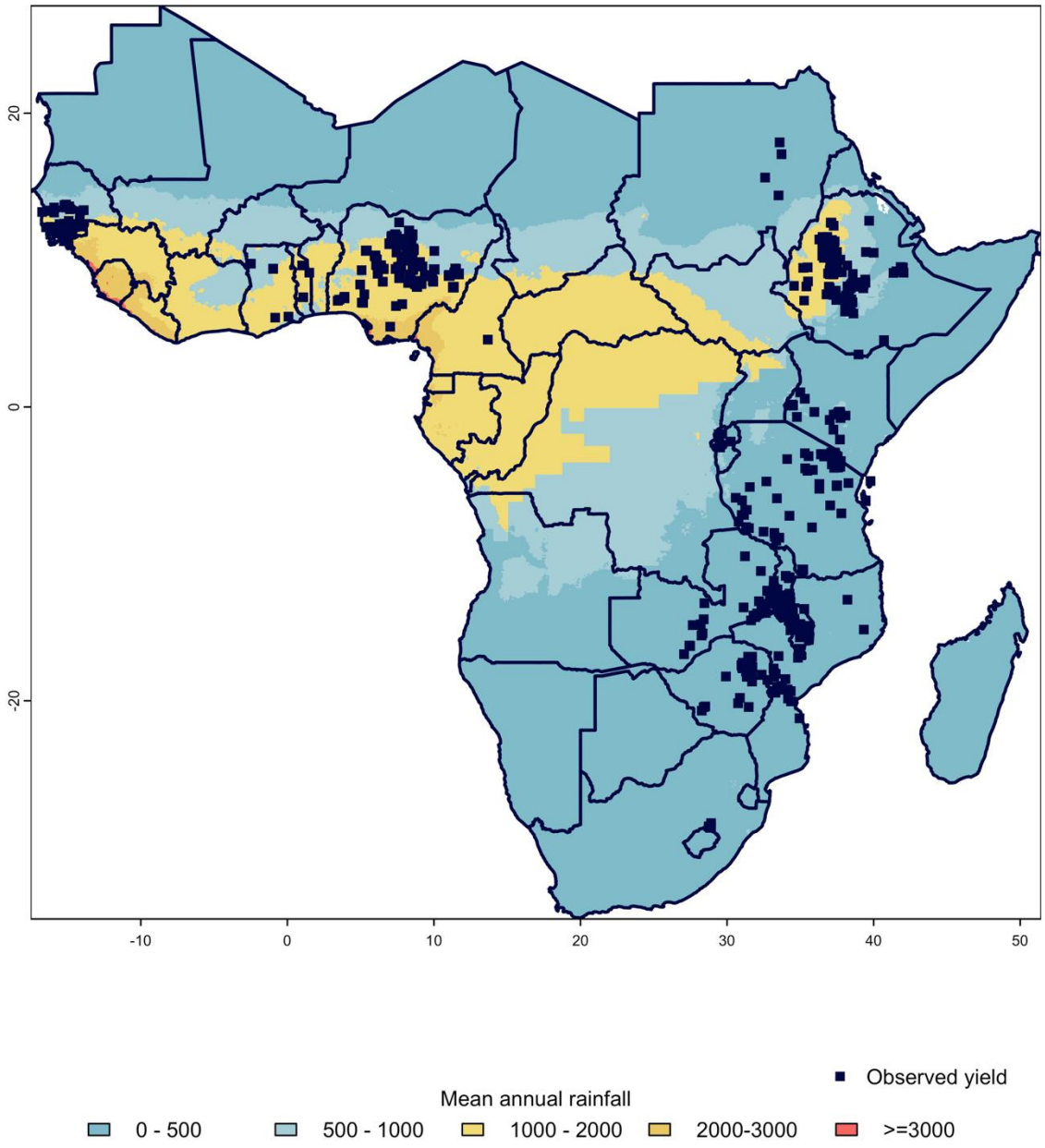

Figure 2:Spatial distribution of experimental maize trial sites across rainfall regimes in Sub-Saharan Africa

Each observation includes grain yield, applied nutrient rates, trial year, and geographic coordinates. Trials with incomplete nutrient information, implausible application rates, or missing location data were excluded. Yield values are standardized to common units prior to analysis. The trial data are used exclusively to estimate conditional yield response surfaces and are not used to infer farmer behavior or adoption patterns.

Maize price data are compiled from national market information systems, FEWS NET, and the World Food Programme, yielding an unbalanced panel of monthly prices from more than 1,800 markets across Sub-Saharan Africa over the period 2007–2024. When prices are missing from all sources for a particular month, the World Bank's monthly food price estimates by product and market (Andrée, 2021) is used to input the missing price data. Market prices are spatially interpolated to continuous surfaces using geographic proximity and market-access covariates, generating pixel-level expected prices and historical price distributions. These distributions

capture local price levels, volatility, and tail behavior, and are used directly in the stochastic simulations.

Retail prices for urea were obtained from observed market-level data collected in 2023 and 2024 across regional markets, sourced from AfricaFertilizer.org (2024). These observations provide recent, location-specific information on fertilizer retail prices and are used to anchor the spatial fertilizer price surfaces and calibrate input cost variability in the analysis.

We derive seasonal rainfall metrics from the Climate Hazards Group InfraRed Precipitation with Stations (CHIRPS) v2.0 dataset, which provides high-resolution precipitation estimates from 1981 onward (Funk et al., 2015). Daily rainfall fields are aggregated to maize-specific growing seasons using spatially explicit calendars (Van Tricht et al., 2023), producing seasonal rainfall totals, interannual variability measures, and standardized rainfall anomalies.

To characterize biophysical constraints, we use gridded soil properties from ISRIC SoilGrids and the Africa Soil Information Service (AfSIS), including soil organic carbon, pH, sand and clay fractions, and effective cation exchange capacity, originally available at 250 m resolution and resampled to the analysis grid (Hengl et al., 2017; Leenaars et al., 2014).

Maize production areas are delineated using the Spatial Production Allocation Model (SPAM), which provides crop-specific harvested area estimates across SSA (IFPRI, 2024). All profitability and risk metrics are computed conditional on maize presence.

To capture crop stress and failure risk, we incorporate vegetation indices and survey-based crop-loss observations. Seasonal NDVI and EVI time series from MODIS are used to characterize vegetation condition and stress responses (Didan, 2021). In addition, we use plot- and household-level survey data collected over two agricultural seasons, comprising approximately 6,000 observations of severe crop loss from the Nigeria LSMS. These data support the estimation of pixel-level crop-loss probability models as functions of rainfall shocks and biophysical conditions.

Table 1:Data sources, coverage, and role in the analysis

Data component	Description	Spatial/temporal coverage	Resolution	Source(s)	Role in analysis
Yield response trial data	Georeferenced maize yield observations under varying NPK rates	SSA; multiple countries and AEZs	Plot-level	CIMMYT & national research systems (various years); legacy fertilizer response trials from the Carob project (Hijiman et al, 2023)	Estimation of spatially explicit maize yield response functions
Seasonal rainfall	Growing-season rainfall totals, variability, and anomalies	SSA, 1981–2024	~5 km	CHIRPS v2.0 (Funk et al., 2015)	Climate input to yield and crop-loss models; rainfall shocks in simulations
Soil & biophysical properties	Organic carbon, pH, sand, clay, effective CEC	SSA (static)	250 m (resampled)	ISRIC SoilGrids (Hengl et al., 2017); AfSIS (Leenaars et al., 2014)	Controls for biophysical heterogeneity in yield and loss models
Maize production area	Spatial distribution of maize cultivation	SSA	~10 km	SPAM 2017/2020 (IFPRI, 2024)	Masking and aggregation of results to relevant maize-growing areas
Maize output prices	Monthly market prices, detrended and seasonally aggregated	~1,850 markets; 2007–2024	Market-level -> gridded	FEWS NET; WFP; FAOSTAT; World Bank; national market systems	Estimation of spatial price surfaces and output price risk
Nitrogen fertilizer prices	Market-level nitrogen fertilizer prices	SSA	Market-level -> gridded	AFA (Africa Fertilizer & Agribusiness Partnership)	Fertilizer cost calculation and net return estimation
Vegetation indices (NDVI/EVI)	Seasonal vegetation condition and anomalies	SSA; 2000–2024	250–500 m	MODIS (Didan, 2015)	Proxy for crop stress and input to crop-loss modeling
Crop loss survey data	Binary indicators of severe crop failure	Selected SSA countries, two seasons	Household/plot-level	Nigerian LSMS data	Estimation of pixel-level crop loss probability models
Accessibility & population	Travel time to markets; population density	SSA (static)	~1 km	GPWv4 (CIESIN, 2018); accessibility surfaces (Nelson et al., 2019)	Market access controls and price interpolation support

4. Results

4.1 Spatial patterns in predicted maize and nitrogen price surfaces

Figure 3 presents spatially explicit maize price surfaces generated from the price prediction model. These surfaces summarize the spatial distribution of expected maize prices and their variability across SSA, providing the market-side inputs to the profitability and risk analysis.

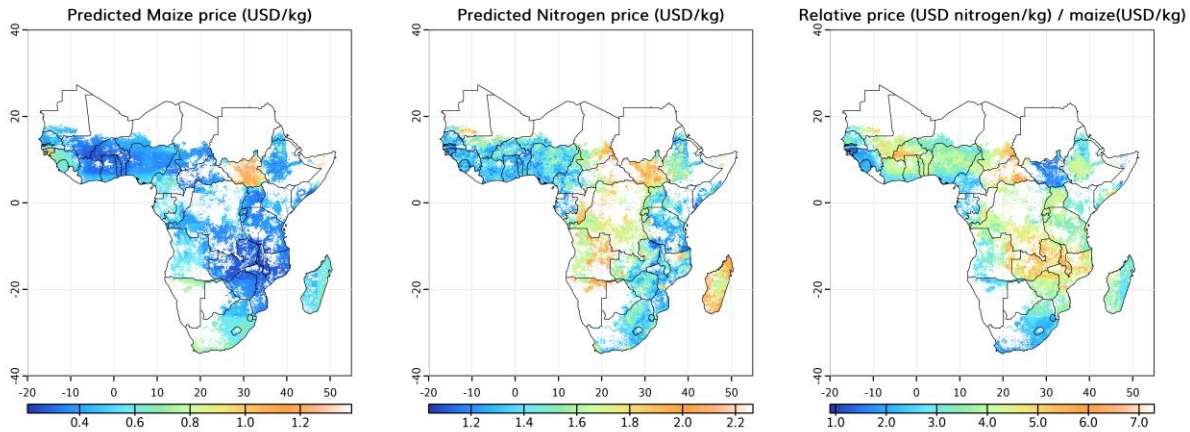

Figure 3: Spatial distribution of predicted maize and fertilizer prices across Sub-Saharan Africa

All three price surfaces exhibit substantial spatial heterogeneity. Predicted maize prices vary widely across the region, with values ranging from approximately 0.4 to 1.2 USD/kg. Lower predicted maize prices dominate across large contiguous areas, while higher prices appear in more spatially concentrated zones.

Predicted nitrogen prices also display pronounced spatial variation, with values ranging from approximately 1.2 to 2.2 USD/kg. The spatial pattern of nitrogen prices differs from that of maize prices, with elevated nitrogen prices observed in several inland regions and comparatively lower prices in other areas. This divergence in spatial structure between input and output prices contributes to variation in their relative magnitudes.

The implied nitrogen-to-maize price ratio shows the greatest degree of spatial dispersion, ranging from approximately 1 to over 7 across the study region. Areas with higher relative price ratios coincide with locations where predicted nitrogen prices are high relative to predicted maize prices, while lower ratios occur where the opposite pattern holds. The spatial distribution of the relative price surface differs from that of maize and nitrogen prices individually, reflecting the combined spatial variation in both input and output prices.

4.2 Spatial heterogeneity in yield response to nitrogen

Figure 4 presents spatially explicit yield response surfaces derived from the maize yield response model, illustrating predicted baseline yields under zero fertilizer application, incremental yield gains from nitrogen use, and implied nitrogen use efficiency (NUE). These surfaces characterize the agronomic component of fertilizer profitability across rainfed maize systems in SSA.

Predicted baseline yields under zero nitrogen application (Figure 4a) show pronounced spatial heterogeneity. Higher baseline yields are concentrated in relatively favorable rainfed systems characterized by adequate growing-season rainfall and more suitable soil conditions, while

substantially lower yields dominate in drier and more marginal production environments. Baseline yield levels provide a critical reference point for interpreting subsequent fertilizer responses, as they influence both absolute production outcomes and the scope for marginal yield gains.

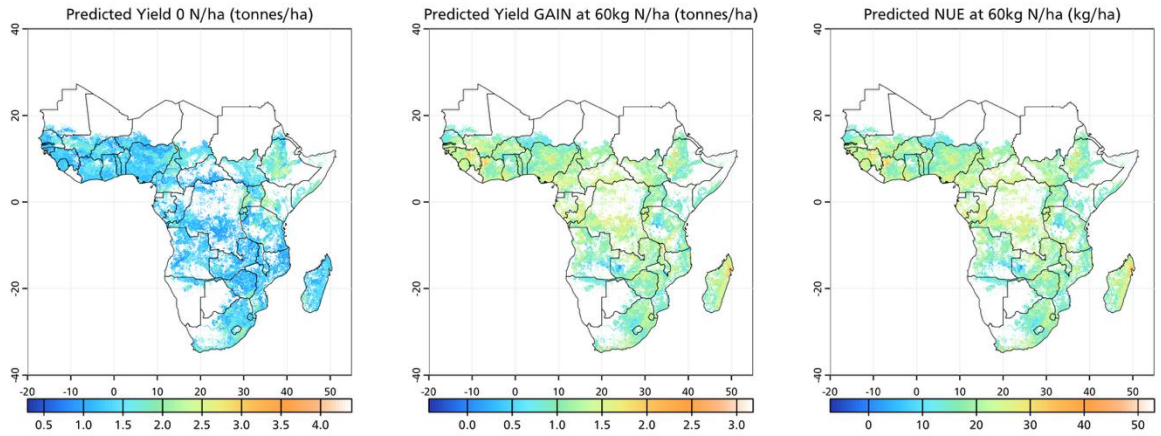

Figure 4:Spatial variation in predicted maize yield response to fertilizer under 60 kg/ha nutrient application rates

Looking at the predicted yield gains from nitrogen application (illustrated at 60 kg N ha^{-1} in Figure 4b), in some locations, nitrogen application is associated with large incremental gains, often exceeding 2–3 t ha^{-1}, whereas in other areas the predicted response is modest or close to zero. Importantly, areas with low baseline yields do not systematically correspond to high nitrogen responsiveness. This finding indicates that yield gaps alone are poor predictors of fertilizer response and that localized soil–climate interactions might play a central role in shaping agronomic returns to nitrogen.

Figure 4c reports the implied nitrogen use efficiency (NUE), defined as yield gain per unit of applied nitrogen. NUE displays similarly strong spatial variation, with high-efficiency zones emerging where rainfall patterns and soil properties allow crops to effectively translate applied nitrogen into grain yield. From an economic perspective, these patterns are particularly relevant, as NUE directly affects the cost-effectiveness of fertilizer investments under prevailing input and output price conditions.

The yield response surfaces shown in Figure 4 demonstrate that agronomic potential is highly heterogeneous across SSA and cannot be adequately summarized using regional averages or coarse agroecological classifications. Baseline productivity, marginal yield response, and nutrient efficiency jointly shape the production-side returns to fertilizer use. These agronomic patterns form the foundation for the stochastic profitability analysis that follows, where yield responses are combined with rainfall variability, price uncertainty, and crop loss probabilities to assess risk-adjusted returns.

4.3 Spatial distribution of predicted crop-loss probability

We first estimate a spatial surface of crop-loss probability that represents baseline vulnerability to production failure under observed biophysical conditions. This crop-loss probability is generated from a supervised learning model trained on NDVI-derived crop-loss indicators and associated covariates capturing rainfall stress and soil constraints. Importantly, this surface is not a Monte Carlo output; it is a first-stage prediction that provides the probabilistic failure risk at each pixel given its local rainfall (and rainfall anomaly) and soil attributes.

Figure 5 maps the predicted crop-loss probability across SSA. The model implies substantial spatial heterogeneity in failure risk, with clusters of elevated probabilities concentrated in locations where rainfall shocks and/or adverse soil conditions are more prevalent. Because

the model is trained on NDVI-based loss signals, this surface reflects historically observed crop condition outcomes, translated into a probabilistic measure that can be combined with economic return simulations.

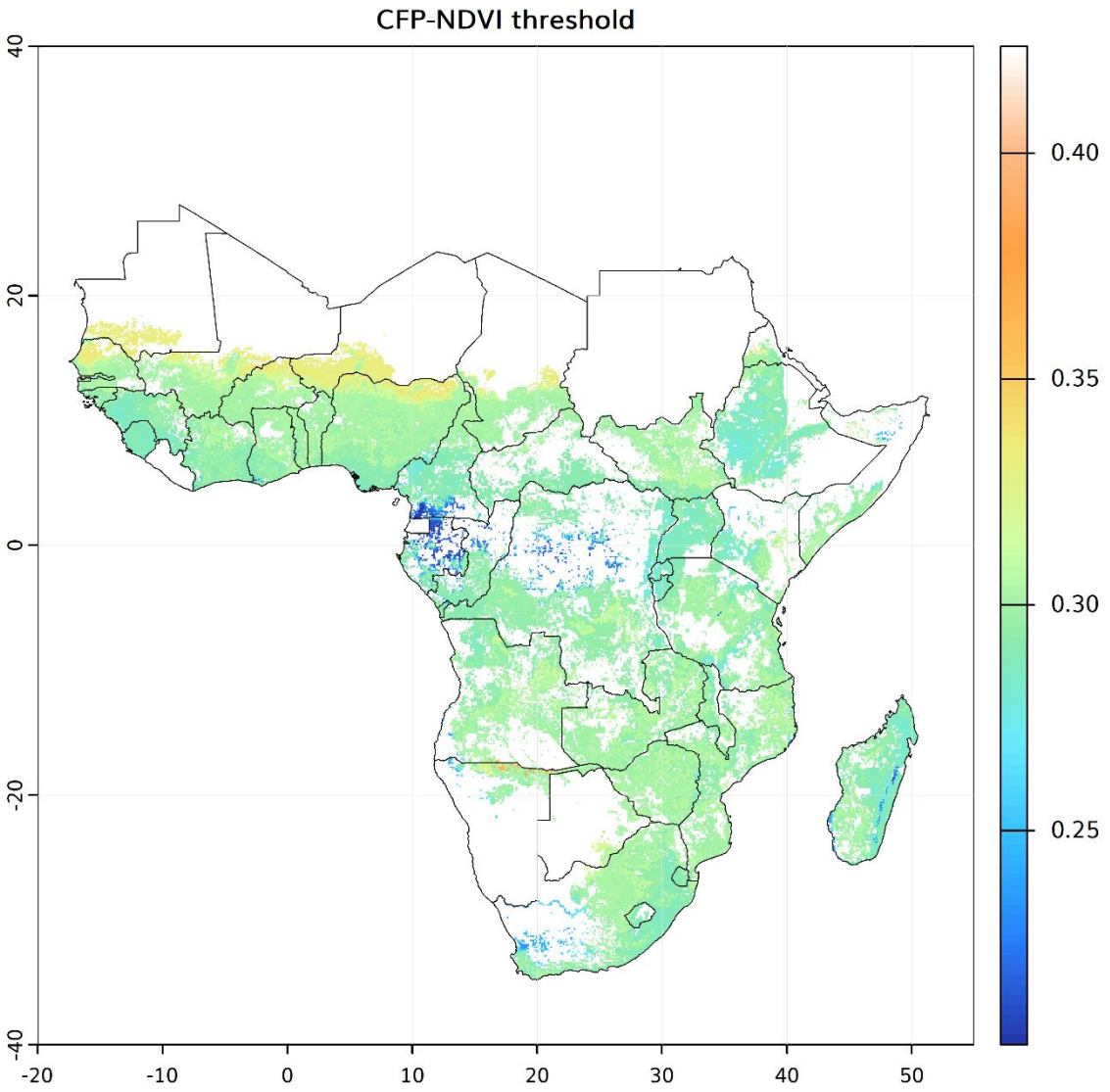

Figure 5:NDVI-derived crop-loss indicators

4.4 Joint spatial dependence from the spatial copula

This section presents empirical evidence on the joint spatial dependence between rainfall and maize prices and evaluates the realism of simulated price surfaces generated by the spatial copula framework. The objective is to assess whether the simulation engine reproduces key spatial and cross-variable features observed in historical data, before embedding these shocks into profitability analysis. The estimated cross-variable dependence between growing-season rainfall and maize prices, summarized using Kendall's τ and mapped to the implied Gaussian copula correlation parameter. Across SSA, the estimated dependence is statistically weak but non-zero, indicating that adverse rainfall outcomes tend to coincide with higher maize prices, though with substantial spatial variation. This dependence is not uniform across space, reflecting heterogeneity in market integration, trade flows, and the extent to which local production shocks are transmitted into prices.

Figures 6 and 7 present examples of simulated maize price and rainfall surfaces generated using the spatial copula framework. Across simulations, the generated surfaces preserve the spatial structure observed in the historical data while introducing stochastic variation across draws. The simulated realizations reproduce the spatial patterns of higher and lower expected maize prices and rainfall observed in the underlying datasets.

Panels comparing simulated and historical moments show close agreement in both the expected values and the coefficient of variation. For maize prices (Figure 6), the spatial distribution of the simulated mean closely matches the observed mean price surface, while the simulated coefficient of variation mirrors the spatial heterogeneity in historical price variability. Similarly, for rainfall (Figure 7), the simulated mean rainfall and coefficient of variation align closely with their observed counterparts, reproducing both the level and spatial distribution of rainfall variability across the study region.

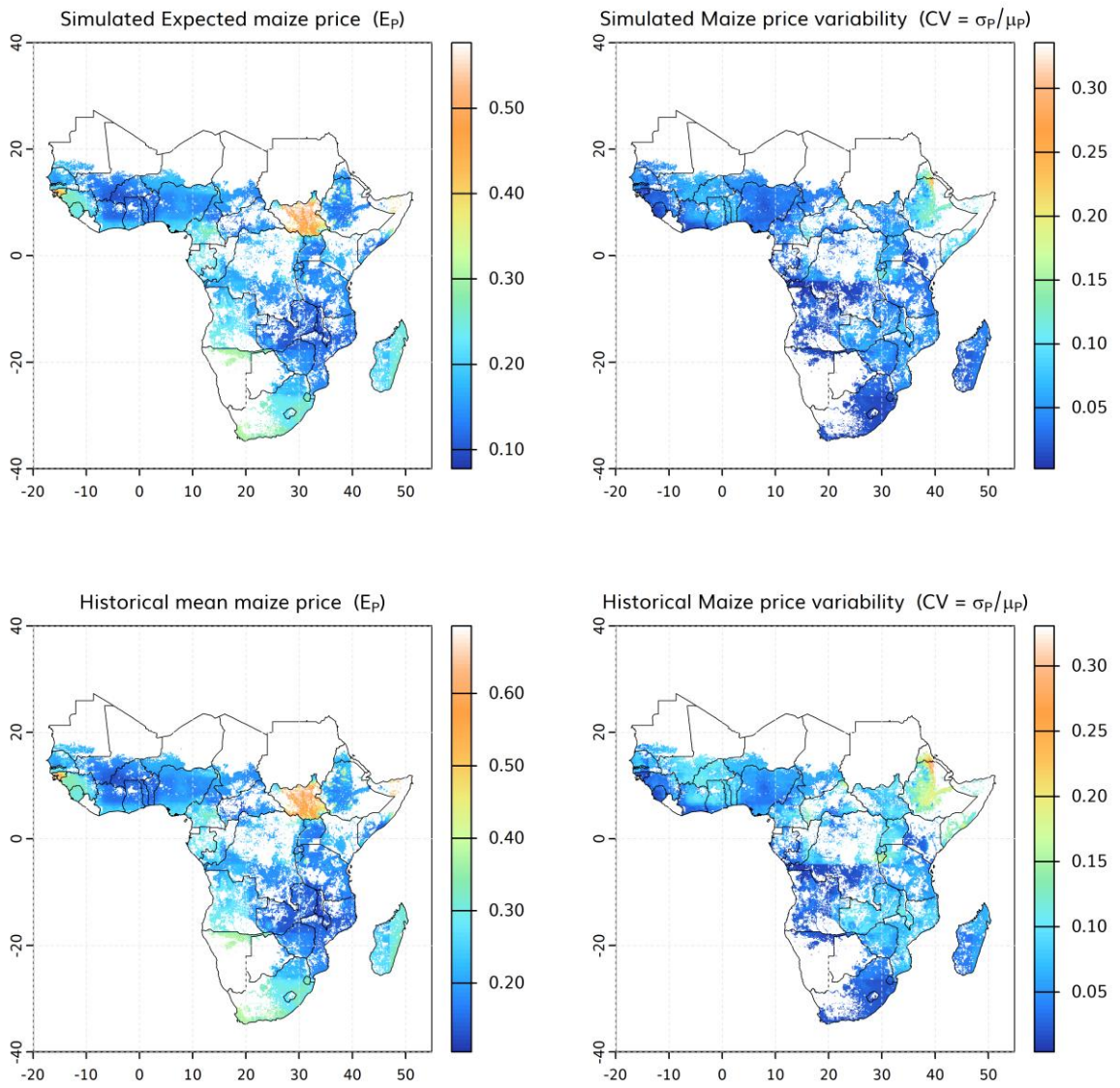

Figure 6: Simulated rainfall and price realizations generated from the spatial copula model, compared with historical distributions

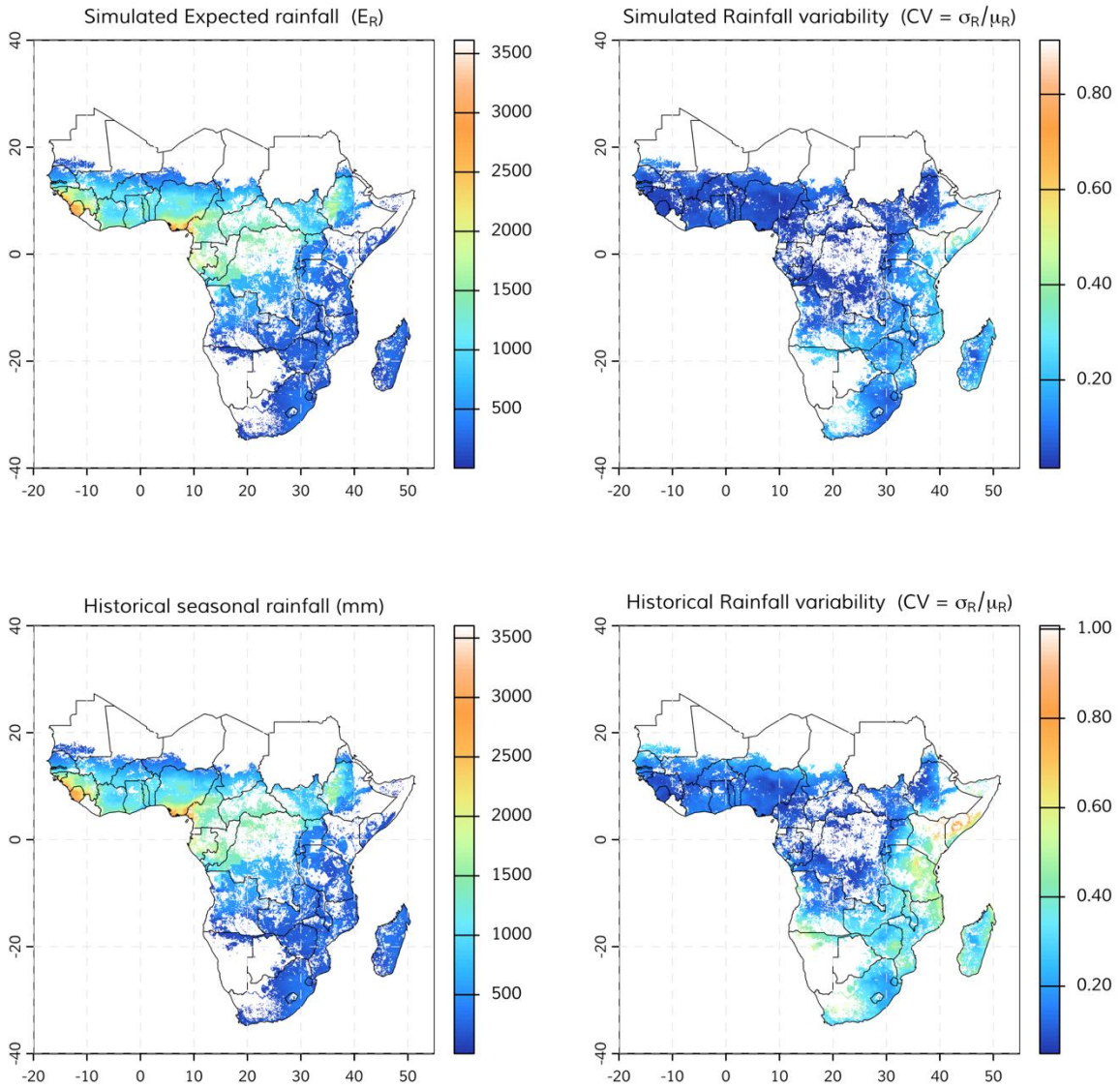

Figure 7Simulated rainfall realizations generated from the spatial copula model, compared with historical distributions

4.5 Probability of loss and downside profit risk

Beyond mean profitability and overall variance, fertilizer adoption decisions in rainfed systems are strongly influenced by downside risk. To characterize this dimension, we summarize simulated profit distributions using two complementary indicators derived from the Monte Carlo simulations: the probability of economic loss and the severity of losses under extreme adverse outcomes.

Figure 8 illustrates the spatial distribution of these downside risk measures. The probability-of-loss surface reports, for each location, the share of simulated realizations in which net returns to fertilizer are negative, given joint uncertainty in yields, prices, and crop failure. Higher values identify areas where fertilizer investments fail more frequently across plausible states of nature. Complementing this measure, the Conditional Value-at-Risk at the 5 percent level (CVaR(5%)) captures the expected net return among the worst-performing five percent of simulated

outcomes. Whereas the probability of loss reflects how often losses occur, CVaR summarizes how severe those losses are when they materialize. Together, these surfaces distinguish locations where downside risk is driven primarily by frequent small losses from those where risk is dominated by rare but economically severe shocks.

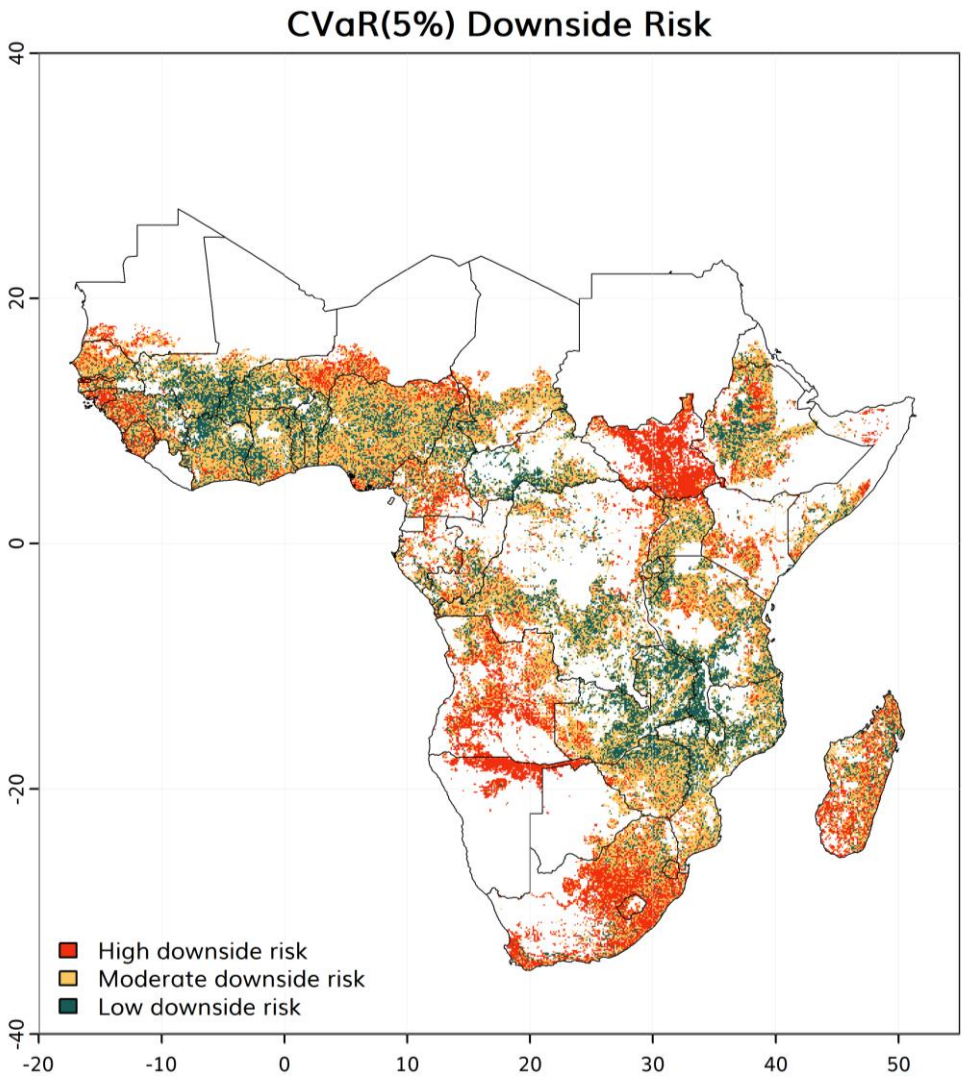

Figure 8:Spatial distribution of downside fertilizer profit risk measured by the 5 percent Conditional Value at Risk

Figure 9 presents the probability of economic loss associated with fertilizer use. The map highlights substantial spatial variation in the frequency with which fertilizer investments generate negative returns across simulated states of nature. In several rainfed regions, the probability of loss exceeds one third of all realizations, indicating that fertilizer investments fail in a large share of plausible seasons even when mean returns are positive.

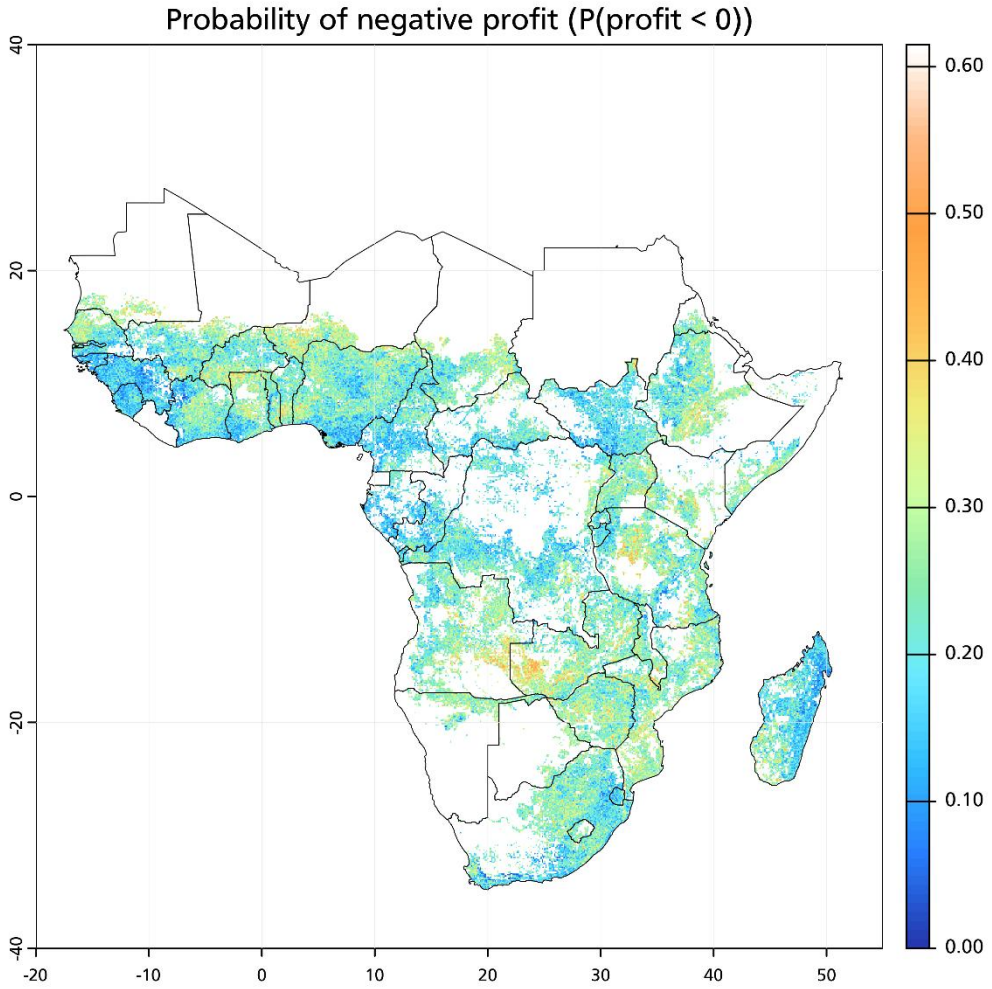

Figure 9:Spatial distribution of the probability of negative fertilizer profits under joint climate and market uncertainty

Taken together, the probability-of-loss and CVaR results demonstrate that fertilizer profitability in SSA is not only spatially heterogeneous but also asymmetrically risky. Regions that appear similar in terms of mean returns often differ sharply in their downside risk profiles. This finding has direct implications for technology targeting, subsidy design, and extension messaging. In low-loss-probability, low-CVaR areas, fertilizer promotion may proceed with limited risk exposure. In contrast, high-loss-probability or high-CVaR zones may require complementary instruments, such as insurance, risk-contingent credit, or stress-tolerant input packages, to mitigate downside risk before intensification can be economically justified.

4.6 Profit–risk quadrants and spatial targeting implications

To synthesize the joint information on expected profitability and risk exposure, we classify simulation outcomes into profit–risk quadrants using pixel-level summaries derived from the Monte Carlo simulations. Specifically, we combine mean simulated profit with measures of variability CV to identify areas characterized by high versus low expected returns and high versus low risk. This classification provides a transparent framework for translating complex stochastic outcomes into decision-relevant spatial categories.

Figure 10 presents the profit–risk quadrant map constructed using median-based thresholds of expected profit and profit variability across the SSA domain. Four distinct regimes emerge. First, areas with high expected profits and low variability represent robust investment zones, where fertilizer use is both profitable on average and relatively stable across climate and price

realizations. These areas are concentrated in regions with favorable rainfall distributions, moderate price exposure, and relatively low crop-loss probabilities. Second, high-profit but high-variability zones exhibit attractive mean returns but substantial downside risk, reflecting strong sensitivity to adverse rainfall or price shocks. In these areas, fertilizer investments may be profitable but require complementary risk-management instruments.

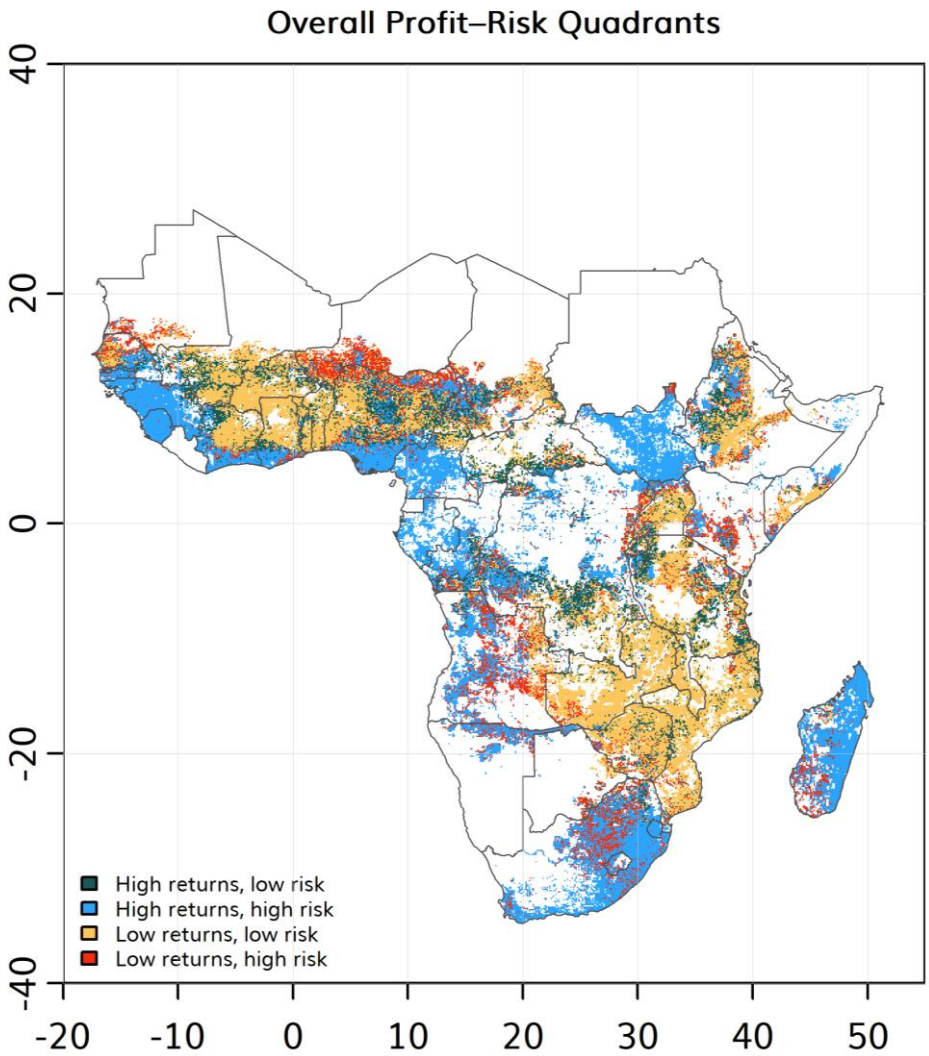

Figure 10: Spatial classification of expected returns and risk using continental thresholds

Third, low-profit and low-variability areas are characterized by consistently modest or negative returns, even under favorable realizations. These zones typically coincide with biophysically constrained environments where yield response to fertilizer is weak, and risk exposure offers limited upside potential. Finally, low-profit and high-variability regions represent the most precarious investment environments, where fertilizer use is both unprofitable on average and highly exposed to risk, including frequent crop-loss events.

Importantly, Figure 11 demonstrates that the spatial configuration of profit–risk quadrants is highly sensitive to the choice of reference thresholds used for classification. When thresholds for expected returns and downside risk are defined at the continental level, locations are ranked relative to a common benchmark across sub-Saharan Africa, preserving cross-country differences in profitability and risk. In contrast, applying country-specific thresholds re-centers the classification within national distributions, leading to substantial reallocation of locations

across quadrants despite unchanged underlying profit distributions. As a result, areas identified

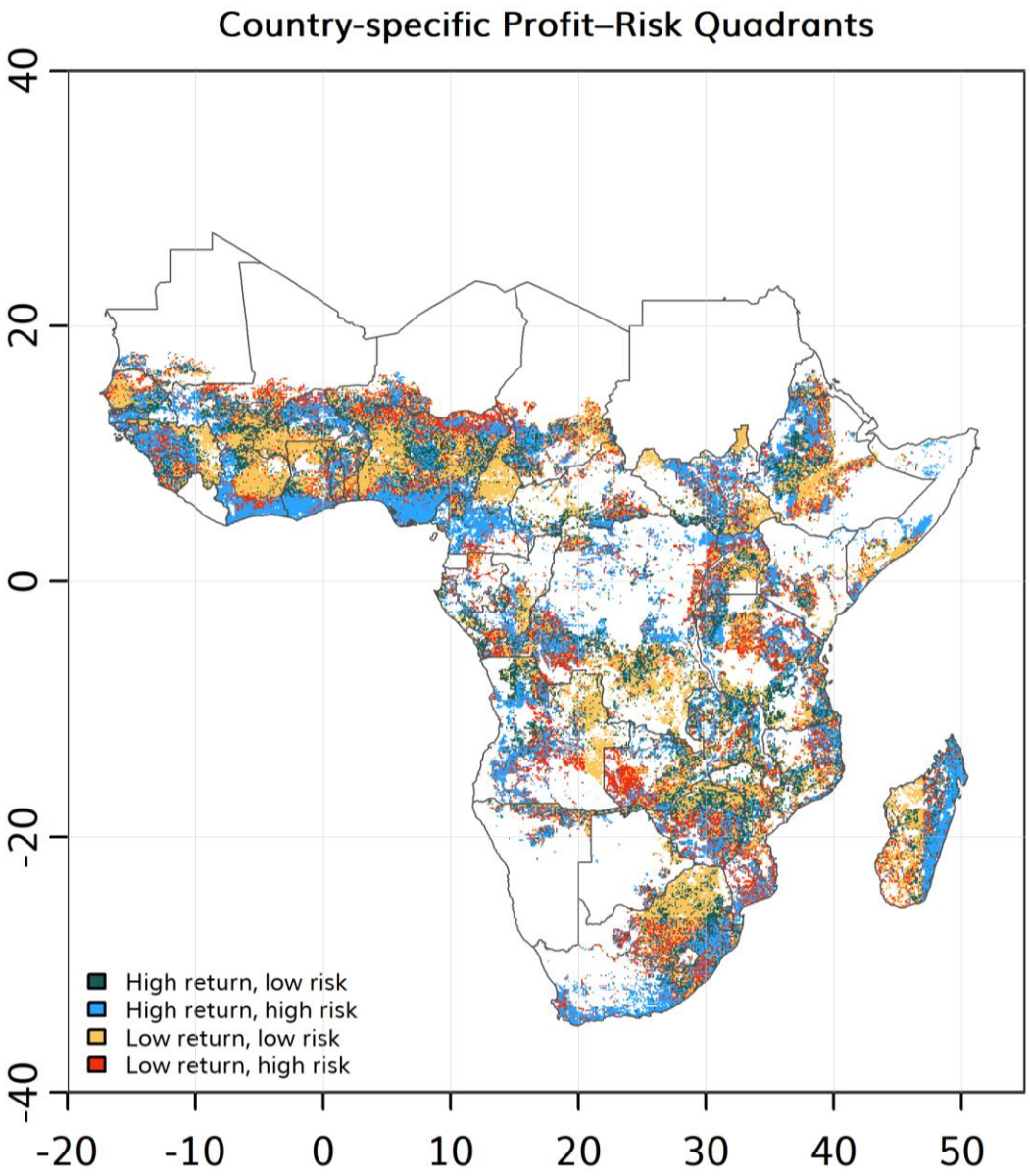

Figure 11:Spatial classification of expected returns and risk using country-specific thresholds

as high-return or high-risk under continental benchmarks may shift categories when evaluated relative to country-level reference points. This reclassification underscores that profit–risk quadrants reflect relative positioning within a chosen comparison set, and cautions that targeting outcomes depend critically on whether thresholds are defined using continental or national reference distributions.

Aggregating quadrant classifications by country and subnational administrative units further reveals substantial within-country heterogeneity (Figure 12). In many cases, countries contain a mix of robust, risky, and unviable zones, underscoring the limitations of national-level recommendations. Even within high-potential maize-growing countries, large shares of cultivated area fall into high-risk or low-return quadrants, implying that spatially uniform fertilizer promotion policies are likely to generate uneven and potentially adverse outcomes.

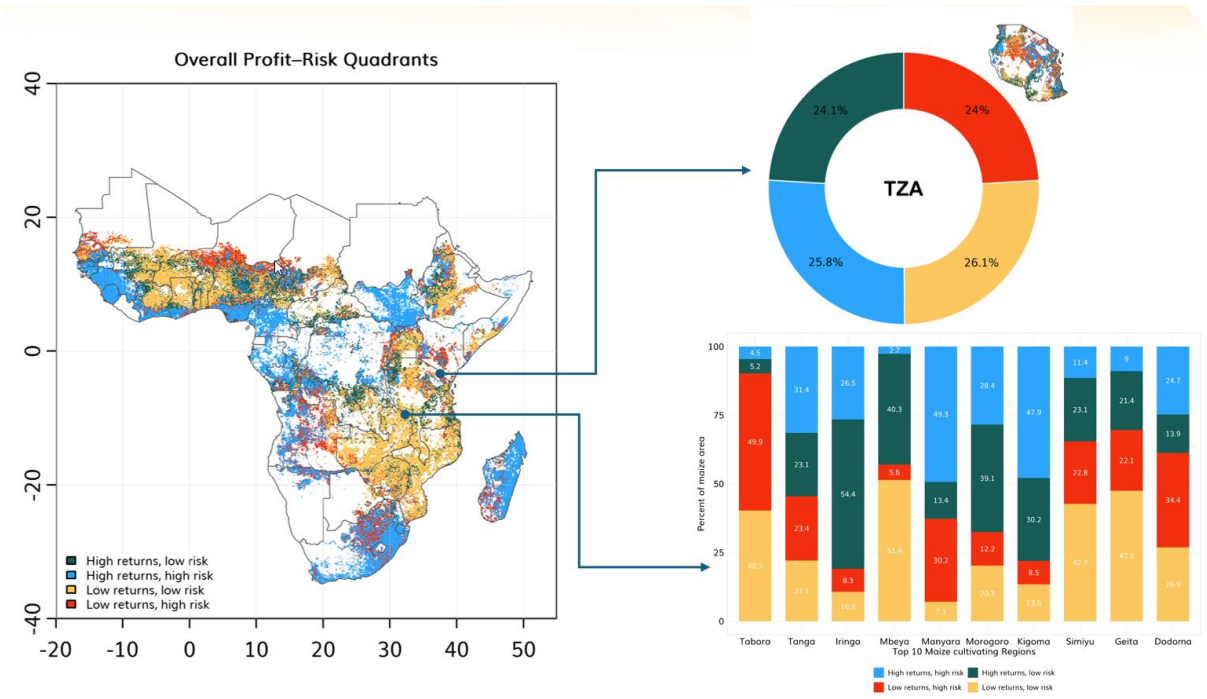

Figure 12: Spatial aggregation of simulated fertilizer returns distributions from continental to subnational scales

The profit–risk quadrant analysis provides a compact summary of the simulated outcomes by jointly characterizing expected returns and exposure to risk, including crop loss. By combining these dimensions, the framework distinguishes locations where fertilizer use is associated with higher expected returns and lower risk from those where potential gains are offset by elevated uncertainty. This classification highlights spatial heterogeneity in the economic conditions under which fertilizer intensification may be viable and supports comparison of targeting outcomes under alternative risk and threshold definitions.

5. Discussions and conclusions

This report provides a spatially explicit economic characterization of fertilizer investments under uncertainty in rainfed maize systems of Sub-Saharan Africa. By integrating predictive models of yield response, rainfall variability, prices, and crop loss within a unified simulation framework, we move beyond average agronomic response analysis to examine the full distribution of economic outcomes faced by smallholder farmers.

A central finding is that expected profitability alone is an inadequate metric for evaluating fertilizer investments. Across large areas, mean returns to fertilizer application are positive, yet the probability of loss and the severity of downside outcomes remain substantial. This result aligns with economic theories of technology adoption under risk and liquidity constraints and helps reconcile the long-standing puzzle of low fertilizer uptake despite apparently favorable average returns reported in agronomic trials.

The explicit quantification of downside risk provides new insight into this disconnect. Metrics such as the probability of loss, Value at Risk (VaR), and Conditional Value at Risk (CVaR) reveal sharp spatial gradients in exposure to extreme negative outcomes. In many locations, CVaR values indicate that when losses occur, they are large relative to farm income, implying potentially catastrophic consequences for risk-averse households. These downside risks are largely invisible in yield-focused or mean-return analyses but are economically central to farmer decision-making.

The spatial copula framework further demonstrates that joint climate–price uncertainty materially amplifies economic risk. Rainfall shocks and price variability are not independent,

and their co-movement increases the dispersion of simulated profits relative to scenarios that consider climate or prices in isolation. This finding highlights the importance of modeling dependence structures explicitly, rather than relying on additive or independent risk assumptions commonly used in *ex ante* analyses.

Importantly, the results reveal substantial spatial heterogeneity in the risk–return trade-off. Locations with similar expected profits can differ markedly in variability, downside risk, and loss probability. The profit–risk quadrant classification illustrates this heterogeneity in a policy-relevant way, identifying areas where fertilizer intensification is both profitable and relatively resilient, as well as areas where high variability or high loss probability dominates potential gains. These patterns persist when results are aggregated to country and sub-national levels, underscoring the value of spatially explicit targeting.

The results of this study are most usefully interpreted through a set of concrete use cases rather than as a single policy prescription. First, the framework can support spatial targeting of fertilizer promotion, helping planners distinguish areas where fertilizer is likely to be profitable under most conditions from those where returns are highly uncertain. By identifying locations where downside risk dominates expected gains, the analysis can help avoid blanket recommendations and reduce the risk of promoting fertilizer use in settings where farmers are likely to experience losses. In this sense, the framework serves as a screening tool that complements existing agronomic suitability assessments by incorporating economic risk

Second, the framework can inform the design and prioritization of complementary interventions where fertilizer responsiveness exists but risk exposure remains high. In such settings, the results can help clarify whether constraints are primarily related to production risk, market volatility, or the risk of catastrophic crop failure. This feature is relevant for choosing between alternative support mechanisms, such as risk-contingent credit, insurance pilots, soil health investments, or market infrastructure improvements. Importantly, the framework does not determine which intervention should be implemented, but it helps narrow the policy space by linking specific risk profiles to reasonable categories of response. Used in this way, the approach provides an evidence base for more context-specific and economically grounded fertilizer programming, aligned with the realities faced by smallholder farmers.

6. Limitations and outlook

Several limitations of the analysis are important to note for interpreting the results and defining potential use cases. First, crop failure risk is modeled indirectly using remotely sensed vegetation indices and biophysical attributes, calibrated with a limited set of survey-based loss observations. While this approach captures broad spatial patterns of crop stress and relative failure risk, it cannot reliably distinguish partial yield reductions from complete crop loss, nor does it capture management-driven or idiosyncratic loss mechanisms. Estimated failure probabilities should therefore be interpreted as relative indicators of exposure to catastrophic risk rather than precise estimates of loss frequencies.

Second, the temporal coverage of crop loss information is limited. Survey-based loss observations currently span only a small number of years, constraining the ability to characterize rare but severe failure events. As a result, extreme downside risk may be understated in environments where infrequent shocks dominate outcomes. Expanding the availability of longitudinal, plot-level loss data across additional countries would substantially improve calibration and validation of the failure component and is a clear priority for future work.

Third, the simulation framework assumes stationarity in historical rainfall distributions. Although the model reproduces observed spatial patterns and interannual variability, it does not incorporate projected shifts in climate regimes. Estimated profit and risk metrics therefore reflect historical conditions and should not be interpreted as forecasts under future climate

change scenarios. Extending the framework to non-stationary climate processes would require additional assumptions and is beyond the scope of the present analysis.

Fourth, maize and fertilizer prices are treated as exogenous with respect to simulated production outcomes. While the spatial copula captures observed price levels, volatility, and co-movement with rainfall, the framework does not model market feedbacks arising from widespread production shocks. In reality, correlated crop failures may induce price responses that amplify realized downside risk, which is not explicitly captured here.

Fifth, the yield response models capture systematic variation associated with rainfall, soils, and nutrient application, but do not explicitly represent other sources of yield uncertainty such as pests, diseases, input quality variation, or farm-level management heterogeneity. These omitted factors may further widen the distribution of realized outcomes relative to those simulated.

Direct validation of simulated profit distributions is inherently constrained by the lack of spatially comprehensive, long-run data on farm-level profits. As a result, the framework should not be interpreted as providing point predictions of profits. Validation is instead conducted at the component level. Yield response functions are estimated from georeferenced trial data and evaluated using out-of-sample performance metrics. Rainfall and price simulations are assessed by their ability to reproduce observed means, variability, and spatial dependence patterns. These checks provide confidence that the simulated inputs are statistically consistent with observed processes, while acknowledging that structural simplifications remain.

Within these limitations, the framework offers a practical basis for future extensions and applied use. As data availability improves, particularly for crop loss and management heterogeneity, the approach can be refined without altering its core structure. More immediately, the framework can be used diagnostically to compare risk profiles across regions, to explore how alternative cost structures or price environments affect profitability, and to assess how different forms of risk mitigation might alter the distribution of returns. The primary contribution of this work is therefore not precise prediction, but a transparent and internally consistent way to organize evidence on fertilizer profitability under uncertainty, supporting more risk-aware analysis and discussion in policy and investment contexts.

References

Abay, K. A., Chamberlin, J., Chivenge, P., & Spielman, D. J. (2025). Fertilizer, soil health, and economic shocks: A synthesis of recent evidence. Food Policy, 133, 102892.

AfricaFertilizer.org. 2024. https://africafertilizer.org/#/en

Ayalew, H., Chamberlin, J. and Newman, C., 2022. Site-specific agronomic information and technology adoption: A field experiment from Ethiopia. *Journal of Development Economics*, 156, p.102788. https://doi.org/10.1016/j.jdeveco.2021.102788

Assefa, B. T., Reidsma, P., Chamberlin, J., & van Ittersum, M. K. (2021). Farm-and community-level factors underlying the profitability of fertiliser usage for Ethiopian smallholder farmers. *Agrekon*, *60*(4), 460-479.

Beaman, L., Karlan, D., Thuysbaert, B., and Udry, C. 2013. Profitability of fertilizer: Experimental evidence from female rice farmers in Mali. *American Economic Review* 103(3): 381-386.

Bonilla-Cedrez, C., Chamberlin, J. and Hijmans, R.J., 2021. Fertilizer and grain prices constrain food production in sub-Saharan Africa. *Nature Food*, *2*(10), pp.766-772. https://doi.org/10.1038/s43016-021-00370-1

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.

Burke, W. J., Frossard, E., Kabwe, S., & Jayne, T. S. (2019). Understanding fertilizer adoption and effectiveness on maize in Zambia. *Food policy*, *86*, 101721. https://doi.org/10.1016/j.foodpol.2019.05.004

Chamberlin, J., & Jayne, T. S. (2013). Unpacking the meaning of “market access”: Evidence from rural Kenya. World Development, 41, 245-264.

CHIRPS v2.0. 2024. Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS). University of California, Santa Barbara, USA. http://chc.ucsb.edu/data/chirps

De Brauw, A., and Eozenou, P. (2014). Measuring risk attitudes among Mozambican farmers. *Journal of Development Economics* 111: 61-74

Dercon, S., and Christiaensen, L. (2011). Consumption risk, technology adoption and poverty traps: Evidence from Ethiopia. *Journal of Development Economics* 96(2): 159-173.

Darko, F.A., Ricker-Gilbert, J., Kilic, T., 2024. Profitability of fertilizer use in SSA evidence from Malawi. World Bank Policy Research Working Paper 10859.

Didan, K. (2021). *MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid V061* [Data set]. NASA Land Processes Distributed Active Archive Center. https://doi.org/10.5067/MODIS/MOD13Q1.061 Date Accessed: 2025-07-21

Druilhe, Z., Barreiro-Hurlé, J., Druilhe, Z., & Barreiro-Hurlé, J. (2012). Fertilizer subsidies in sub-Saharan Africa. ESA Working Paper No. 12–04. https://doi.org/10.22004/AG.ECON.288997

Emerick, K., de Janvry, A., Sadoulet, E., and Dar, M. H. (2016). Technological innovations, downside risk, and the modernization of agriculture. *American Economic Review* 106(6): 1537–1561.

FEWS NET. (2023). *Price and market data for food security analysis*. FEWS NET. Avaliable at Markets and Trade | FEWS NET

Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A. and Michaelsen, J., 2015. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Scientific data*, *2*(1), pp.1-21. 150066 (2015). https://doi.org/10.1038/sdata.2015.66

Gollin, D. and Udry, C., 2021. Heterogeneity, measurement error, and misallocation: Evidence from African agriculture. *Journal of Political Economy*, *129*(1), pp.1-80. https://doi.org/10.1086/711369

Hijmans, R.J., C. Bonilla Cedrez, E.G. Bendito, E. Ochieng, R. Murakami, C. Ngakou Sotakoutsing, A. Sila, S. Mkuhlani. 2023. Carob: A Framework to Standardize Agricultural Data. Excellence in Agronomy Initiative, CGIAR. https://hdl.handle.net/10568/137580

Hijmans, Robert J., Márcia Barbosa, Aniruddha Ghosh, and Alex Mandel. 2024. Package ‘geodata’ for R. https://doi.org/10.32614/CRAN.package.geodata

Holden, S.T., 2018. Fertilizer and sustainable intensification in Sub-Saharan Africa. *Global Food Security*, *18*, pp.20-26.

International Food Policy Research Institute (IFPRI). (2024). Global Spatially-Disaggregated Crop Production Statistics Data for 2020 Version 2.0. *Geospatial Data*. Harvard Dataverse. Retrieved from https://dataverse.harvard.edu/citation?persistentId=doi:10.7910/DVN/SWPENT

Center For International Earth Science Information Network-CIESIN-Columbia University. (2017). *Gridded Population of the World, Version 4 (GPWv4): Population Density, Revision 11* (Version 4.11) [Data set]. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). https://doi.org/10.7927/H49C6VHW Date Accessed: 2025-12-22

Jayne, T. S., Mason, N. M., Burke, W. J., & Ariga, J. (2018). Taking stock of Africa's second-generation agricultural input subsidy programs. *Food Policy*, *75*, 1-14.

Krah, K., Michelson, H., Perge, E., & Jindal, R. (2019). Constraints to adopting soil fertility management practices in Malawi: A choice experiment approach. *World Development*, *124*, 104651. Karlan, D., Osei, R., Osei-Akoto, I., and Udry, C. (2014). Agricultural decisions after relaxing credit and risk constraints. *The Quarterly Journal of Economics* 129(2): 597-652.

Koussoubé, E., & Nauges, C. (2017). Returns to fertiliser use: Does it pay enough? Some new evidence from Sub-Saharan Africa. *European Review of Agricultural Economics*, *44*(2), 183-210.

Leenaars, J. G. B., Kempen, B., van Oostrum, A. J. M., & Batjes, N. H. (2014). Africa Soil Profiles Database: A compilation of georeferenced and standardised legacy soil profile data for Sub-Saharan Africa. *Arrouays et al.(eds.), 2014b*, 51-57.

Le Cotty, T., Maître d'Hôtel, E., Soubeyran, R. and Subervie, J., 2018. Linking risk aversion, time preference and fertiliser use in Burkina Faso. *The Journal of Development Studies*, *54*(11), pp.1991-2006.

Liverpool-Tasie, L. S., Omonona, B. T., Sanou, A., & Ogunleye, W. O. (2017). Maize farming and fertilizers: Not a profitable mix in Nigeria. *Agriculture in Africa: Telling myths from facts*, 105-114.

Maertens, M., Oyinbo, O., Abdoulaye, T. and Chamberlin, J., 2023. Sustainable maize intensification through site-specific nutrient management advice: Experimental evidence from Nigeria. *Food Policy*, *121*, p.102546. https://doi.org/10.1016/j.foodpol.2023.102546

McCullough, E.B., Quinn, J.D. and Simons, A.M., 2022. Profitability of climate-smart soil fertility investment varies widely across sub-Saharan Africa. *Nature Food*, *3*(4), pp.275-285. https://doi.org/10.1038/s43016-022-00493-z

Mobarak, A. M., and Rosenzweig, M.R. (2013). Informal risk sharing, index insurance, and risk taking in developing countries. *American Economic Review* 103(3): 375-380.

Nauges, C., Bougherara, D. and Koussoubé, E., 2021. Fertilizer use and risk: New evidence from Sub-Saharan Africa. TSE Working Papers 21-1266, Toulouse School of Economics (TSE). https://publications.ut-capitole.fr/id/eprint/44008/1/wp_tse_1266.pdf

Nelson, A., Weiss, D. J., van Etten, J., Cattaneo, A., McMenomy, T. S., & Koo, J. (2019). A suite of global accessibility indicators. *Scientific data*, *6*(1), 266.

Oyinbo, O., Chamberlin, J., Abdoulaye, T. and Maertens, M., 2022. Digital extension, price risk, and farm performance: experimental evidence from Nigeria. *American Journal of Agricultural Economics*, *104*(2), pp.831-852. https://doi.org/10.1111/ajae.12242

Palmas, S. and Chamberlin, J., 2020. Fertilizer profitability for smallholder maize farmers in Tanzania: A spatially-explicit *ex ante* analysis. *PloS one*, *15*(9), p.e0239149. https://doi.org/10.1371/journal.pone.0239149

Ragasa, C., Takeshima, H., Asante, S., Amare, M., Ma, N., Olanrewaju, O., & Duchoslav, J. (2025). Maize yield responsiveness and profitability of fertilizer: New survey evidence from six African countries. *Food Policy*, *133*, 102815.

Scrucca, L. (2013). GA: A Package for Genetic Algorithms in R. Journal of Statistical Software, 53(4), 1–37. DOI: 10.18637/jss.v053.i04

Suri, T. and Udry, C., 2022. Agricultural technology in Africa. *Journal of Economic Perspectives*, *36*(1), pp.33-56. https://doi.org/10.1257/jep.36.1.33

Suri, T., 2011. Selection and comparative advantage in technology adoption. *Econometrica*, *79*(1), pp.159-209.

Van Tricht, K., Degerickx, J., Gilliams, S., Zanaga, D., Battude, M., Grosu, A., Brombacher, J., Lesiv, M., Bayas, J. C. L., Karanam, S., Fritz, S., Becker-Reshef, I., Franch, B., Mollà-Bononad, B., Boogaard, H., Pratihast, A. K., Koetz, B., and Szantoi, Z. (2023): WorldCereal: a dynamic open-source system for global-scale, seasonal, and reproducible crop and irrigation mapping, Earth Syst. Sci. Data, 15, 5491–5515, https://doi.org/10.5194/essd-15-5491-2023.

World Food Programme. 2024. https://www.wfp.org/market-analysis

Appendix

Figures 12 and 13shows model-agnostic explanations of the crop failure and yield response models, showing global variable importance, marginal effects through partial dependence and individual conditional expectation curves, and local feature contributions for representative locations, used to assess whether the learned relationships are agronomically and economically plausible..

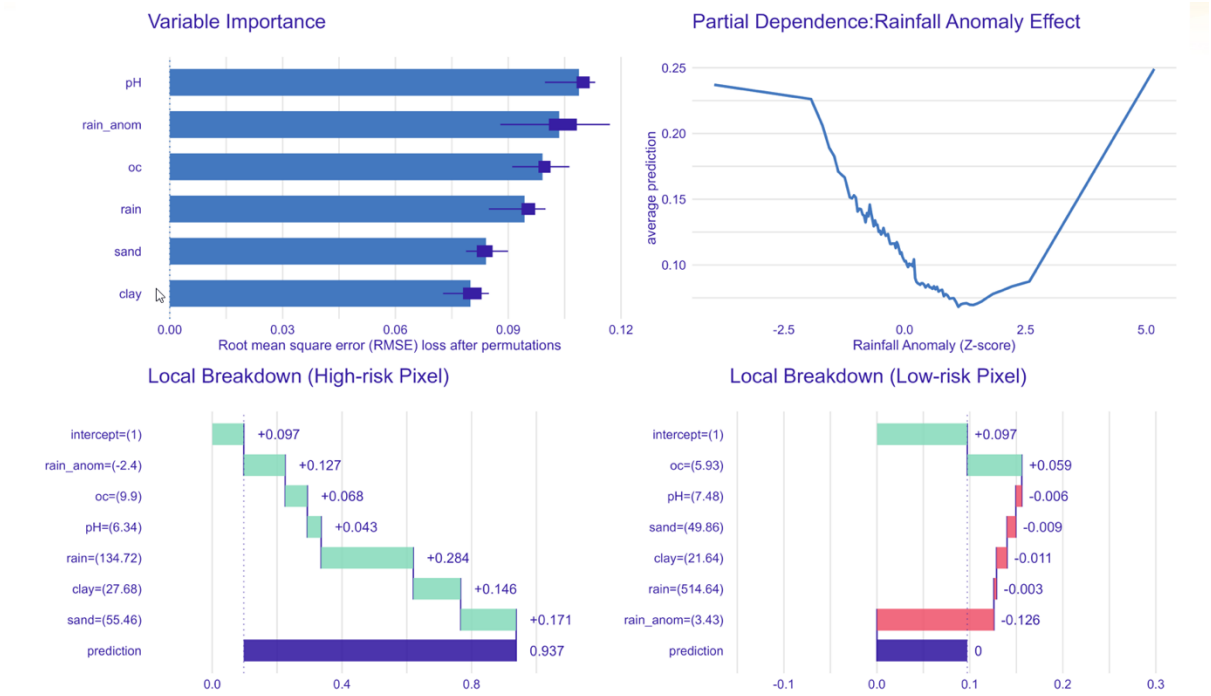

Figure 13: Model-agnostic diagnostics of the crop failure probability model, showing variable importance, marginal rainfall effects, and local explanations

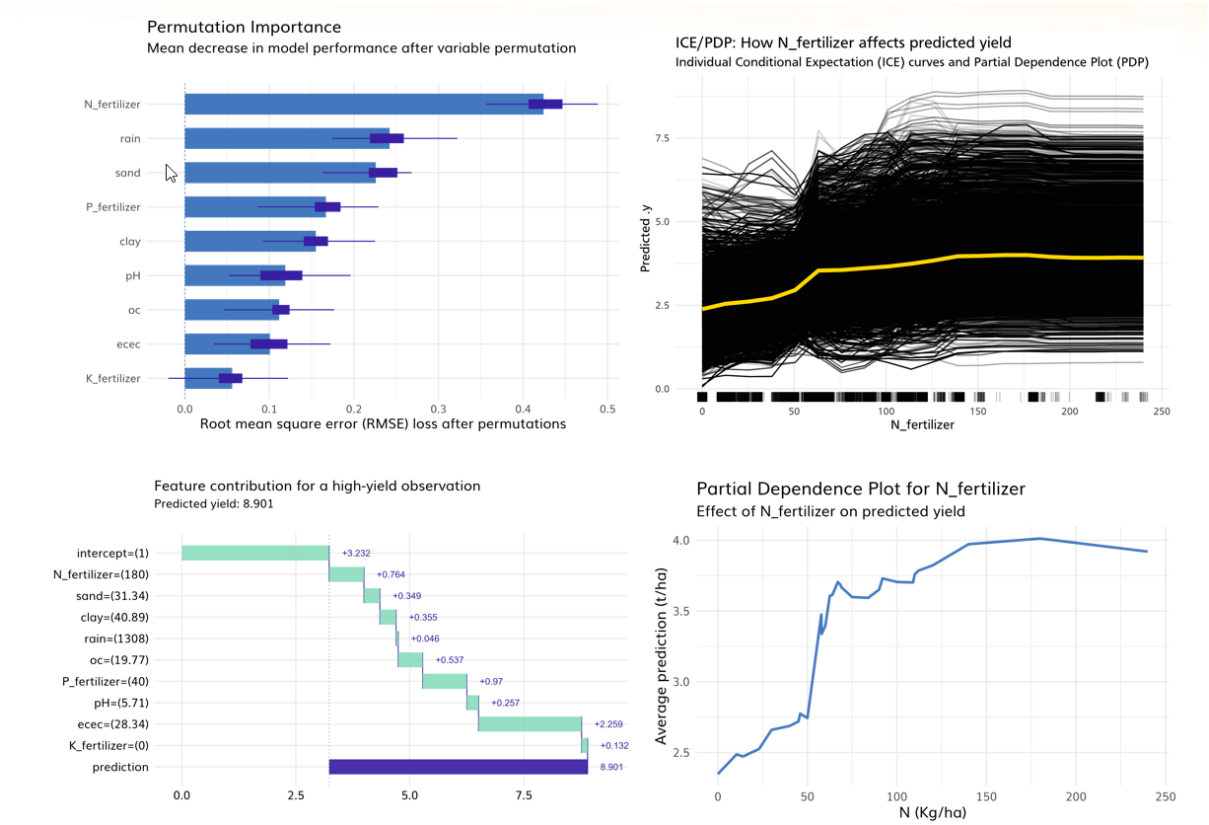

Figure 14:Model-agnostic diagnostics of maize yield response models, including global importance, marginal nitrogen effects, and local contributions

CGIAR

SCIENCE FOR A FOOD-SECURE FUTURE